

# Task C-I: Deepfake Subject Detection

## Semi-Supervised Learning with ECAPA-TDNN (10% Labeled Data)

### 1. Problem Statement

The objective of **Task C-I** is to predict the **identity of the subject being impersonated** in a deepfake audio clip. The model is trained exclusively on *real speech* and evaluated solely on *synthetically generated deepfake audio*.

**Constraints:**

- Only **10% labeled** training data
- **90% unlabeled** data for semi-supervised learning
- Test set contains **only deepfake audio**

This setting requires learning speaker-discriminative representations that generalize robustly across synthetic speech domains.

### 2. Model Architecture

#### Backbone: ECAPA-TDNN

We employ **ECAPA-TDNN** as the speaker embedding extractor due to its strong performance in speaker verification and robustness to channel and acoustic variability.

- **Input Features:** 80-dimensional Mel-frequency cepstral coefficients (MFCCs)
- **Temporal Modeling:** Three SE-Res2Net blocks using dilated 1D convolutions to capture multi-scale temporal context
- **Channel Attention:** Squeeze-and-Excitation (SE) modules emphasize speaker-relevant frequency bands
- **Feature Aggregation:** Multi-layer Feature Aggregation (MFA) concatenates outputs from all Res2Net blocks
- **Pooling:** Attentive Statistics Pooling (ASP) computes weighted means and standard deviations over time
- **Output Head:** A fully connected layer produces fixed-dimensional speaker embeddings

**Embedding Dimension:** 192

#### Classifier and Loss

Speaker embeddings are classified using a linear classifier trained with **Additive Angular Margin Softmax (AAM-Softmax)**, which enforces angular separation between speaker identities:

$$\mathcal{L}_{\text{AAM}} = -\log \frac{e^{s(\cos(\theta_y + m))}}{e^{s(\cos(\theta_y + m))} + \sum_{j \neq y} e^{s \cos(\theta_j)}}$$

*AAM-Softmax improves inter-speaker separation and intra-speaker compactness, which is critical for generalization from real speech to deepfake audio.*

### 3. Methodology

#### Data Preparation

A total of 94 real audio recordings (approximately 1 minute each) are segmented into 2010 **3-second clips** following the task guidelines.

#### Training-Time Augmentation

To enhance robustness against deepfake artifacts, 15 augmentation variants are applied, including:

- Speed and pitch perturbation
- Additive noise (10–25 dB SNR)
- Reverberation using synthetic room impulse responses (RIR)
- Codec degradation and Mu-law companding

The resulting training dataset consists of:

- **Total:** 32,160 clips
- **Labeled:** 3,216 clips (Train: 3,216 / Validation: 483)
- **Unlabeled:** 28,944 clips

#### Semi-Supervised Learning Framework

To leverage unlabeled data, we adopt an iterative self-training strategy with a cold-start retraining mechanism. Let  $\mathcal{D}_L$  and  $\mathcal{D}_U$  denote labeled and unlabeled datasets, respectively.

1. **Warm-up:** Train an initial model  $M_0$  using only  $\mathcal{D}_L$
2. **Pseudo-Labeling:** Predict labels for samples in  $\mathcal{D}_U$  using  $M_k$
3. **Filtering:** Retain high-confidence predictions where  $p(\hat{y}|x) > \tau$ , forming  $\hat{\mathcal{D}}_U$
4. **Retraining:** Train a new model  $M_{k+1}$  *from scratch* on  $\mathcal{D}_L \cup \hat{\mathcal{D}}_U$

Steps 2–4 are repeated until convergence or a maximum of  $K$  iterations.

#### Key Hyperparameters:

- Confidence threshold:  $\tau = 0.95$
- Retraining from scratch mitigates confirmation bias and error accumulation

#### Test-Time Augmentation

At inference, multiple perturbed versions of each test sample are evaluated. Predictions are aggregated via probability averaging, and samples with low agreement are flagged as unstable.

Table 1: Training Hyperparameters

| Parameter            | Value                                              |
|----------------------|----------------------------------------------------|
| Batch Size           | 32                                                 |
| Learning Rate        | 0.001                                              |
| Epochs               | 20                                                 |
| Confidence Threshold | 0.95                                               |
| Max Hypotheses       | 3                                                  |
| Optimizer            | Adam ( $\text{weight\_decay} = 2 \times 10^{-5}$ ) |
| LR Scheduler         | StepLR ( $\text{step\_size} = 5, \gamma = 0.5$ )   |

## 4. Observations

- Naïve pseudo-labeling without augmentation led to poor initial performance
- Heavy augmentation significantly improved robustness to deepfake artifacts
- Test-time augmentation reduced prediction variance and improved confidence calibration
- F1 score improved from  $\sim 33.64\%$  to  $\sim 65.70\%$  on the test set

## 5. Challenges and Solutions

| Challenge                  | Solution                                       |
|----------------------------|------------------------------------------------|
| Limited labeled data       | Semi-supervised self-training                  |
| Pseudo-label noise         | Confidence filtering and cold-start retraining |
| Domain shift to deep-fakes | Heavy augmentation and TTA                     |